

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

	Employee Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

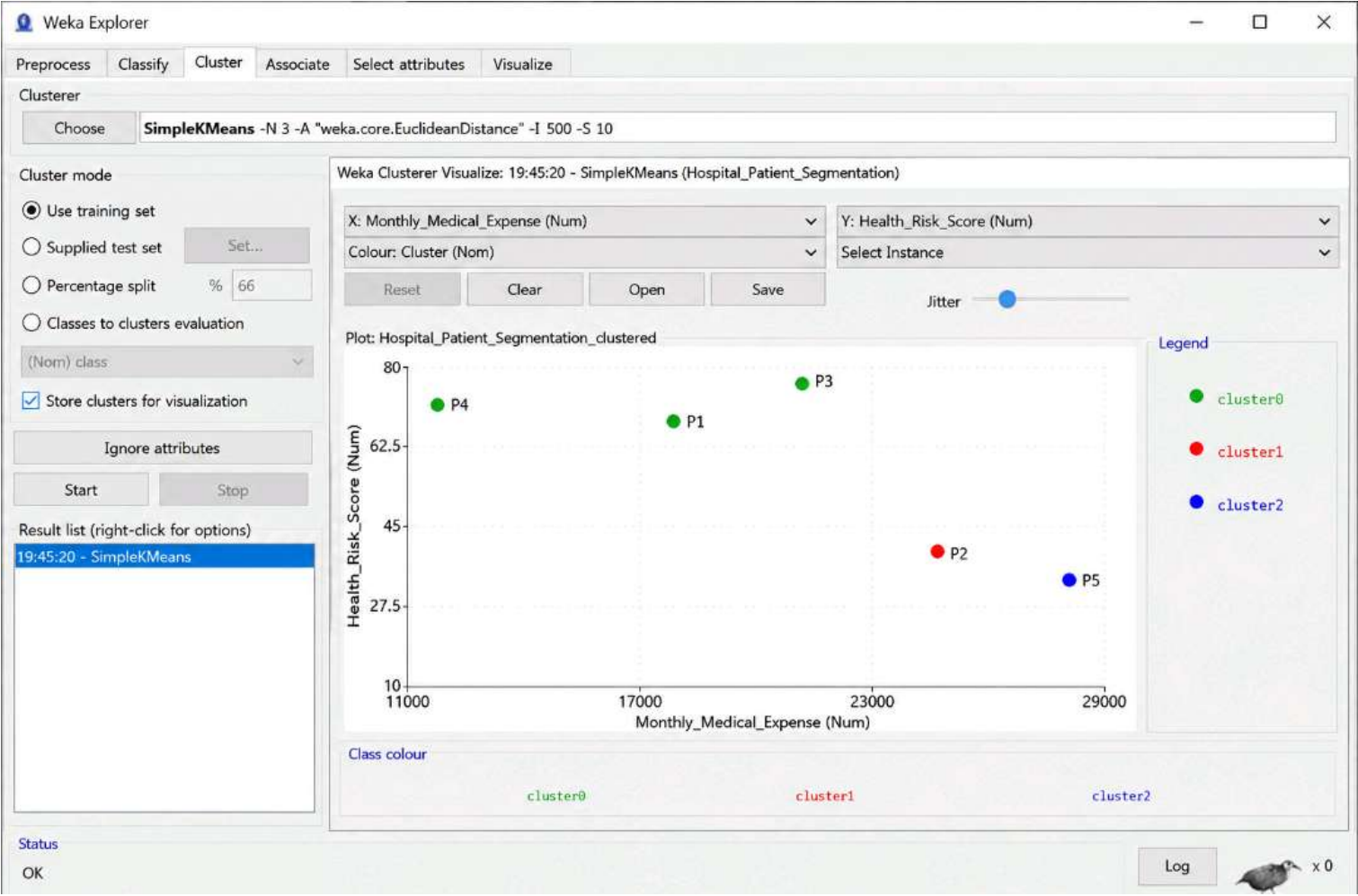
K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.





Clusterer

Choose HierarchicalClusterer -N 2 -L SINGLE -B -V -S EUCLIDEAN

Cluster mode

☒ Use training set☐ Supplied test set

Set...

☐ Percentage split

% 66

☐ Classes to clusters evaluation

(Nom) class

☒ Store clusters for visualization

Ignore attributes

Start

Stop

Result list (right-click for options)

20:15:47 - HierarchicalClusterer

Weka Clusterer Visualize: 20:15:47 - HierarchicalClusterer (E_Commerce_Order_Analysis)

X: Instance_number (Num)

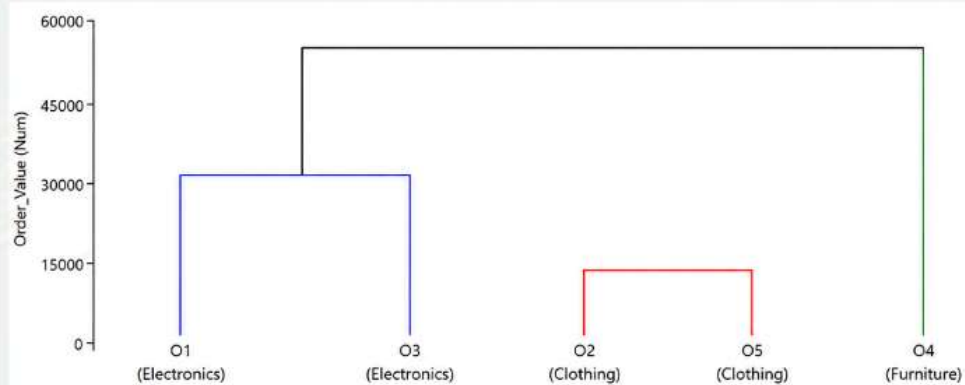
Y: Order_Value (Num)

Colour: Product_Category (Nom)

Select Instance

Jitter

Dendrogram



Output

```
=== Run information ===
Scheme:      HierarchicalClusterer -N2 -L SINGLE -B -V -S EUCLIDEAN
Relation:    E_Commerce_Order_Analysis
Instances:   5
Attributes:  2

=== Clustering model (full training set) ===
Cluster 1:  01, 03
Cluster 2:  02, 05
Cluster 3:  04
```

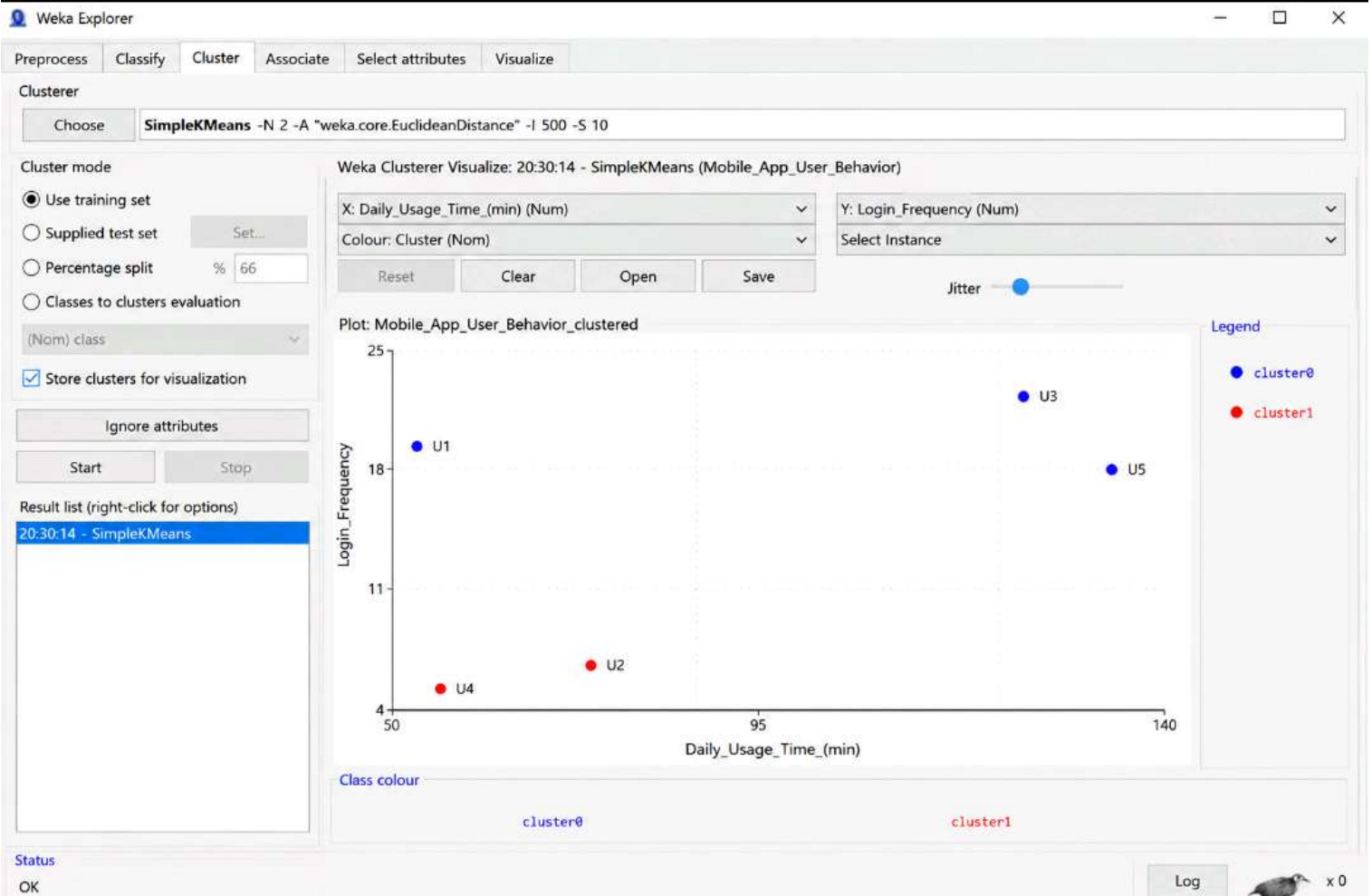
Status

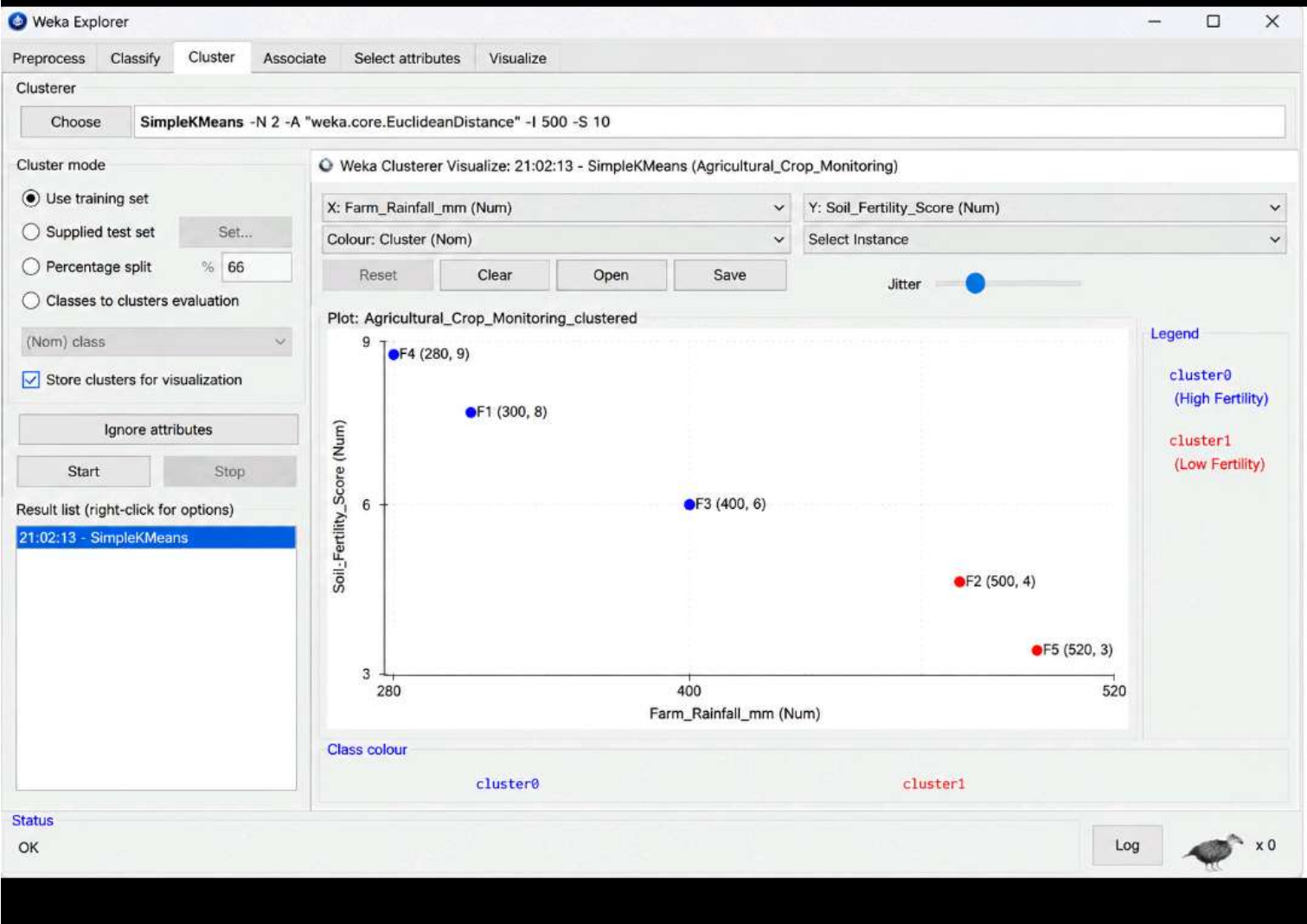
OK

Log



x 0





Weka Explorer

PreprocessClusterAssociateSelect attributesVisualize

Clusterer

ChooseSimpleKMeans -N 3 -A "weka.core.EuclideanDistance -Rfirst-last" -I 500 -S 10

Cluster mode

Use training set

Supplied test setSet...

Percentage split%66

Classes to clusters evaluation(Nom) class

Start

Stop

Result list (right-click for options)

14:23:51 - SimpleKMeans (Agricultural)

14:34:02 - SimpleKMeans (Employee)

Clusterer output

=== Clustering on Agricultural Crop Monitoring Dataset (Normalized) ===

Number of iterations: 3

Within cluster sum of squared errors: 2.7421

Missing values globally replaced with mean/mode

Cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (2.0)	Cluster# 2 (1.0)
Farm Rainfall (mm)	400.000	290.000	450.000	520.000
Soil Fertility Score	6.000	8.500	5.000	3.000
Crop Yield (tons)	78.000	87.500	75.000	65.000

Cluster assignments:

F1 -> cluster 0

F2 -> cluster 1

F3 -> cluster 1

F4 -> cluster 0

F5 -> cluster 2

=====

=== Clustering on Employee Performance Evaluation Dataset (Normalized) ===

Number of iterations: 2

Within cluster sum of squared errors: 1.8924

Missing values globally replaced with mean/mode

Cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (2.0)	Cluster# 2 (1.0)
Productivity Score	75.200	90.500	60.000	78.000
Attendance (%)	83.200	94.000	70.000	88.000
Skill Assessment Score	74.400	87.000	63.000	72.000

Cluster assignments:

E1 -> cluster 2

E2 -> cluster 1

E3 -> cluster 2

E4 -> cluster 1

E5 -> cluster 0

Status

OK

Log

x 0

Weka Explorer

PreprocessClusterAssociateSelect attributesVisualize

Clusterer

ChooseSimpleKMeans -N 5 -A "weka.core.EuclideanDistance -Rfirst-last" -I 500 -S 10

Cluster mode

Use training set

Supplied test setSet...

Percentage split% 66

Classes to clusters evaluation

(Nom) class

Store clusters for visualization

Start

Stop

Result list (right-click for options)

14:20:11 - SimpleKMeans (Agricultural)

14:22:45 - SimpleKMeans (Employee)

14:26:30 - SimpleKMeans (K=5)

Clusterer output

=== Clustering on Agricultural Crop Monitoring Dataset (Normalized) with K=5 ===

Number of iterations: 3

Within cluster sum of squared errors: 1.8427

Missing values globally replaced with mean/mode

Cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (1.0)	Cluster# 2 (1.0)	Cluster# 3 (1.0)	Cluster# 4 (1.0)
Farm Rainfall (mm)	400.000	290.000	300.000	400.000	500.000	520.000
Soil Fertility Score	6.000	8.500	8.000	6.000	4.000	3.000
Crop Yield (tons)	78.000	87.500	85.000	80.000	70.000	65.000

Cluster assignments:

F1 -> cluster 1

F2 -> cluster 3

F3 -> cluster 2

F4 -> cluster 0

F5 -> cluster 4

=====

=== Clustering on Employee Performance Evaluation Dataset (Normalized) with K=5 ===

Number of iterations: 2

Within cluster sum of squared errors: 1.2764

Missing values globally replaced with mean/mode

Cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (1.0)	Cluster# 1 (2.0)	Cluster# 2 (1.0)	Cluster# 3 (0.0)	Cluster# 4 (1.0)
Productivity Score	75.200	93.000	58.500	78.000	0.000	88.000
Attendance (%)	83.200	96.000	70.000	88.000	0.000	92.000
Skill Assessment Score	74.400	90.000	63.000	72.000	0.000	84.000

Cluster assignments:

E1 -> cluster 4

E2 -> cluster 1

E3 -> cluster 2

E4 -> cluster 1

E5 -> cluster 0

Status

OK

Log

x 0